2.1 Transformations of Quadratic Functions

Name _____ Period ____ Date _____

Algebra II 2.1 (30-minute plan) · Student sheet

I can…

- describe transformations of $f(x) = x^2$.
- find the vertex of a transformed quadratic function and graph it.
- write rules for quadratic functions after transformations.

Vocabulary · fill in the missing word

Quadratic function — $f(x) =$ _____, where $a \neq 0$

Parabola — the U-shaped graph of a _____ function

Vertex form — $f(x) =$ _____

Vertex — (h, k), the lowest or _____ point

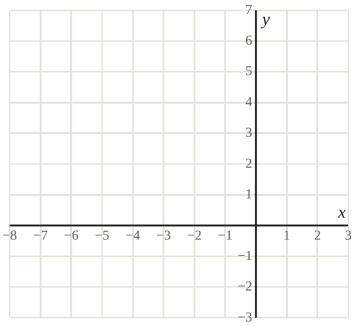
Part 1 · Notes and examples (fill in each note, then work the example)

Vertex form: $f(x) = a(x - h)^2 + k$ has vertex _____.

Translations: $h > 0$ moves the graph _____; $k > 0$ moves it _____.

Example 1

Describe the transformation of $f(x) = x^2$ represented by $g(x) = (x + 4)^2 - 2$. Then graph g.



The value of a: $a < 0$: the parabola _____. $|a| > 1$: _____.

Example 2

Describe the transformation of $f(x) = x^2$ represented by $g(x) = -\frac{1}{2}x^2 + 3$. Give the vertex.

Part 2 · Partner practice

Partner practice

Describe the transformation of $f(x) = x^2$ represented by $g(x) = 2(x - 3)^2$. Give the vertex.